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Python Spatial Stack



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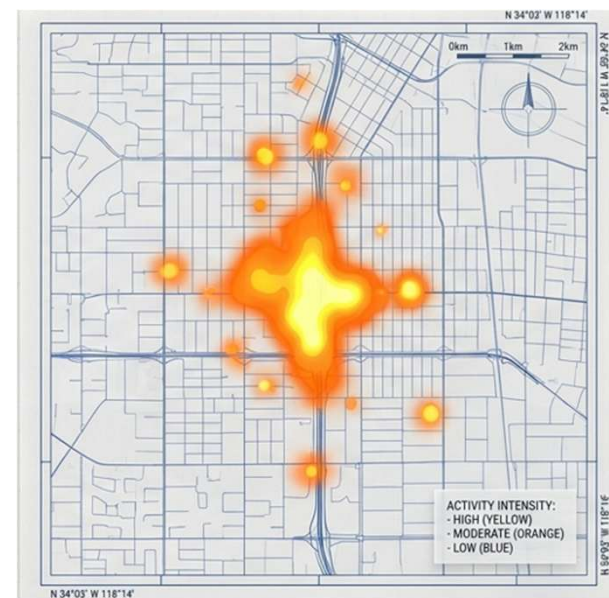
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- Flat reporting obscures non-linear spatial relationships, proximity and clustering

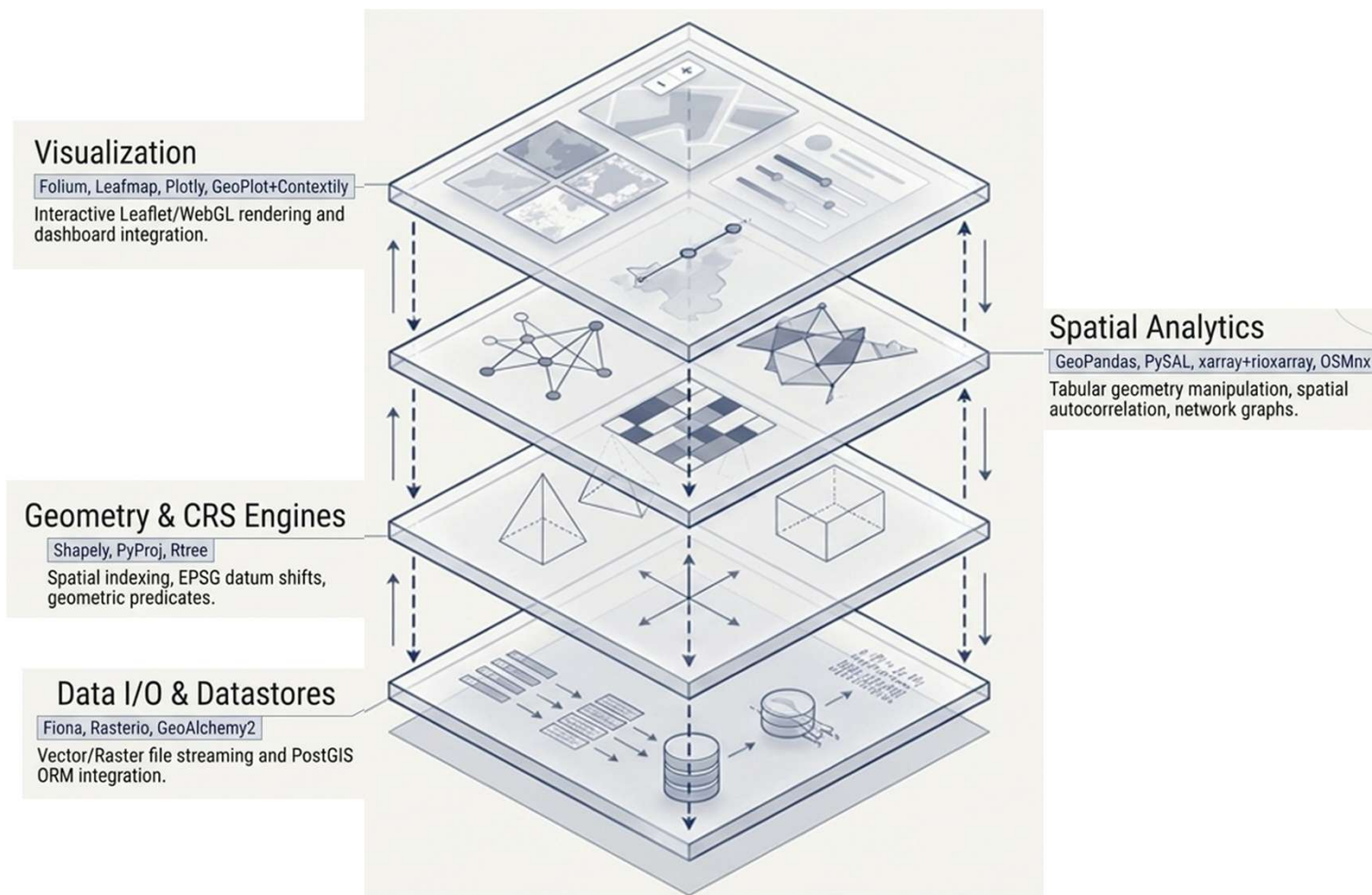
[illegible]

THE GEOGRAPHIC IMPERATIVE

- Spatial visualization anchors raw tabular figures within physical space, instantly unlocking pattern recognition

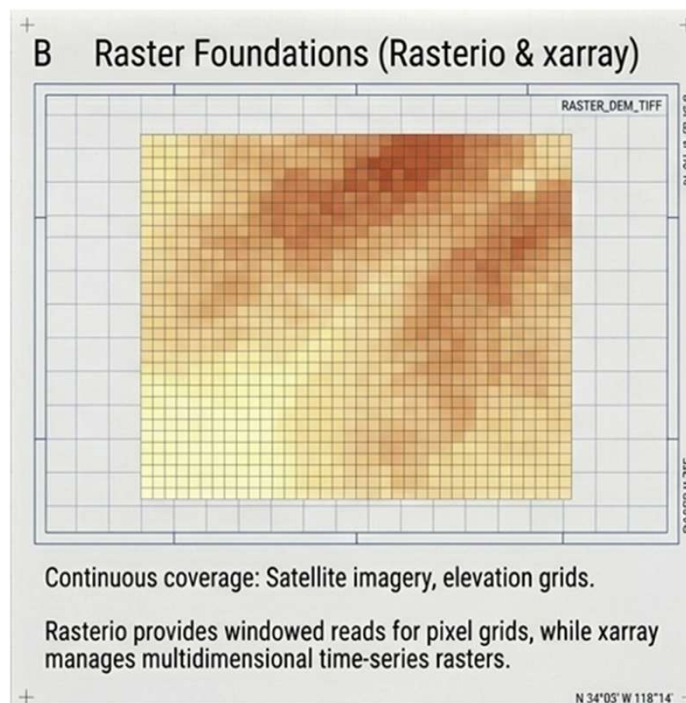
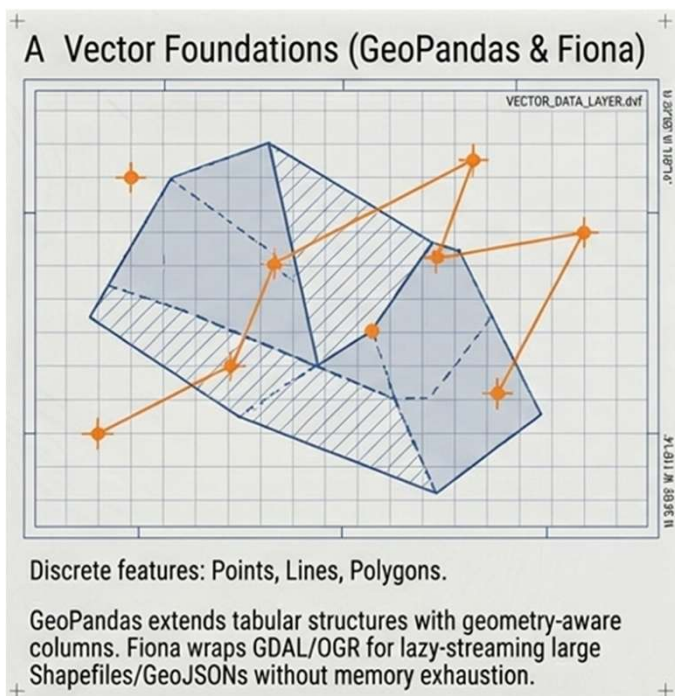


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Geospatial data foundations

- The building blocks of spatial analysis: vector and raster



Geospatial visualization libraries

- A comparative matrix of geospatial rendering engines and capabilities

Folium	PyDeck	Plotly Express	GeoPandas Native
Engine: Leaflet.js	Engine: Deck.gl	Engine: Mapbox GL JS	Engine: Matplotlib
Scale Limit: ~10 ⁴ points (CPU bound)	Scale Limit: ~10 ⁶ + points (GPU bound)	Scale Limit: ~10 ⁵ points (Browser bound)	Scale Limit: Local RAM bound
Dimensionality: 2D Orthogonal	Dimensionality: 2D / 3D Perspective	Dimensionality: 2D Orthogonal	Dimensionality: 2D Static
Best Use Case: Analytical Dashboards & GIS	Best Use Case: Large-scale 3D Point Clouds	Best Use Case: Multi-chart BI Dashboards	Best Use Case: Exploratory Analysis & Print Reports

Folium: python to leaflet.js transformation

Python Environment

```
m = folium.Map(location=[45, -122],  
tiles='CartoDB Positron')
```

Object-Oriented Python.
No HTML5, CSS3, or JS required.

Transformation
& generation

Browser Execution

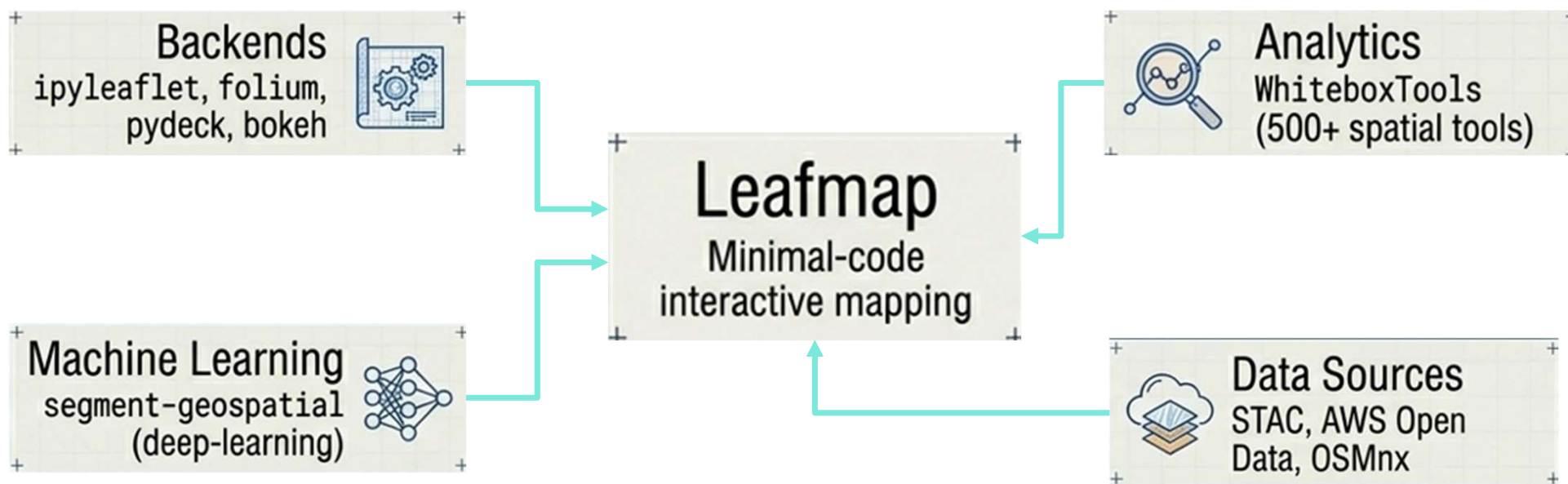


Folium generates an HTML wrapper around Leaflet.js instructions, executing deep zooming, panning, and vector rendering natively in the browser.



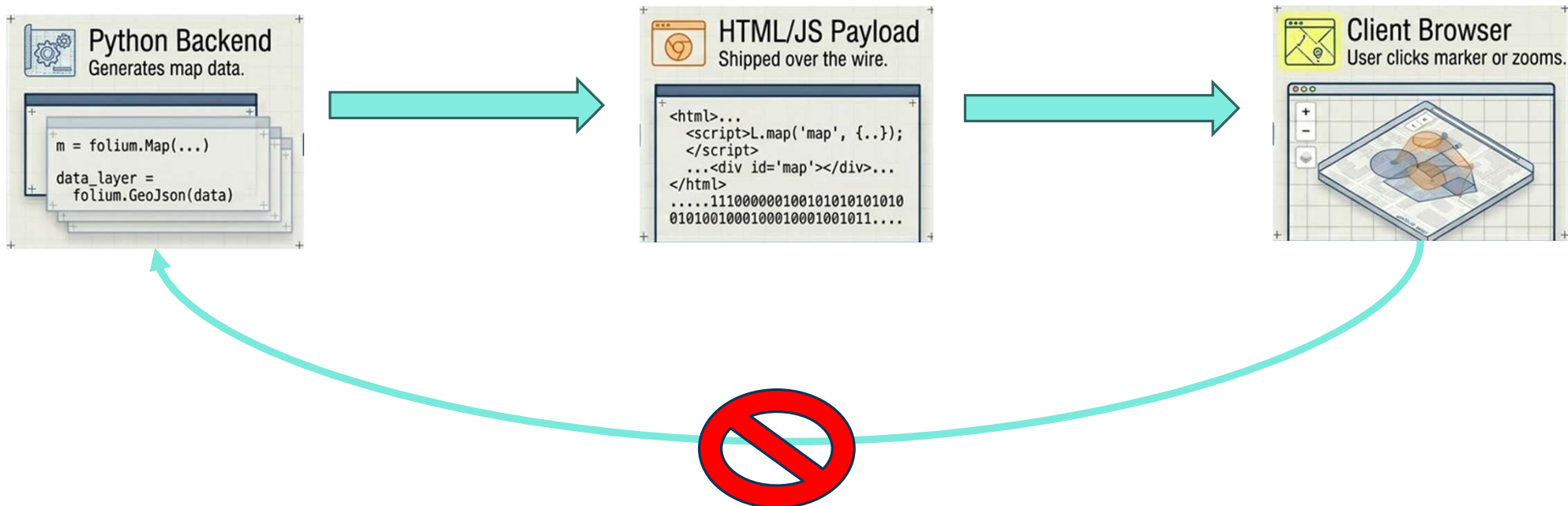
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Leafmap: the unified geospatial HUB

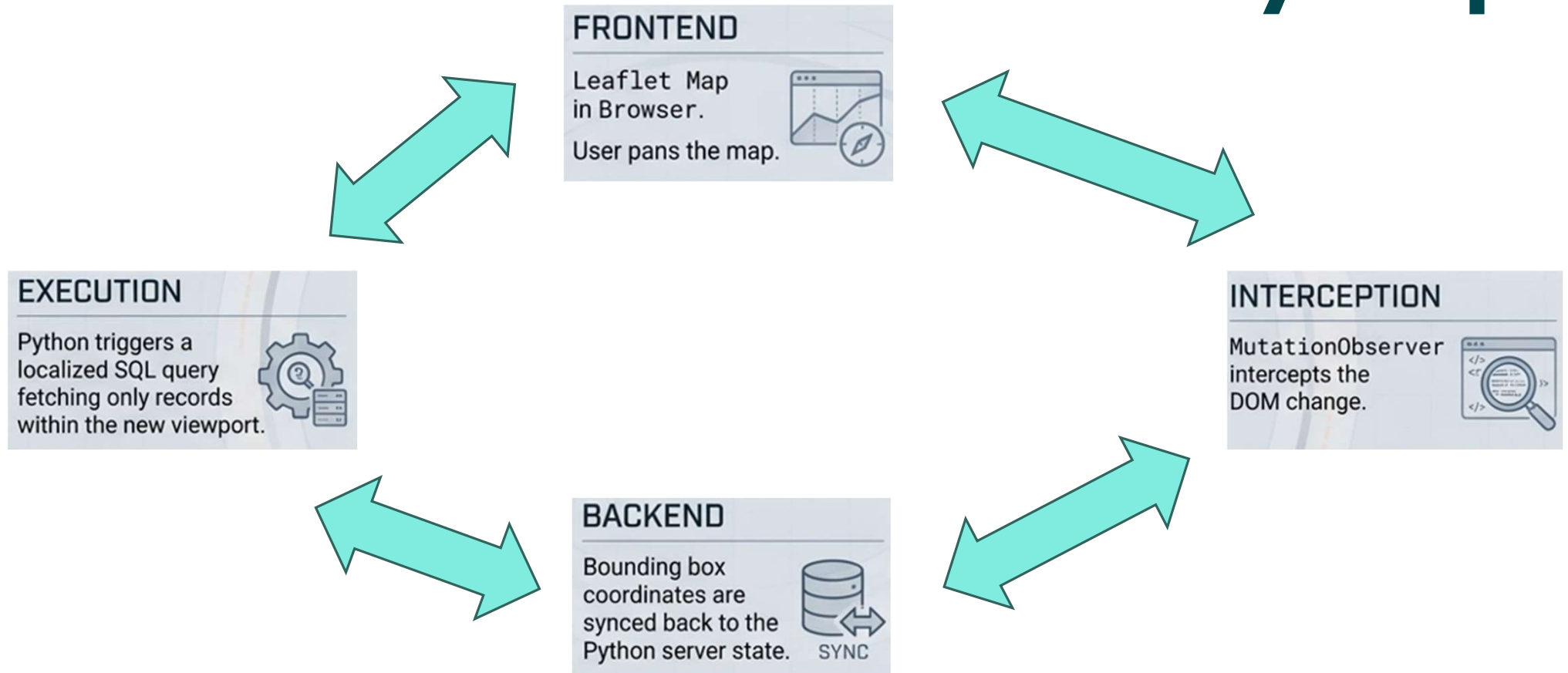


The unidirectional problem: Folium's one-way flow

- Folium acts as a pure rendering engine. Once the HTML payload hits the client, the execution link breaks, Interactions remain isolated in browser memory, failing to trigger backend spatial queries.

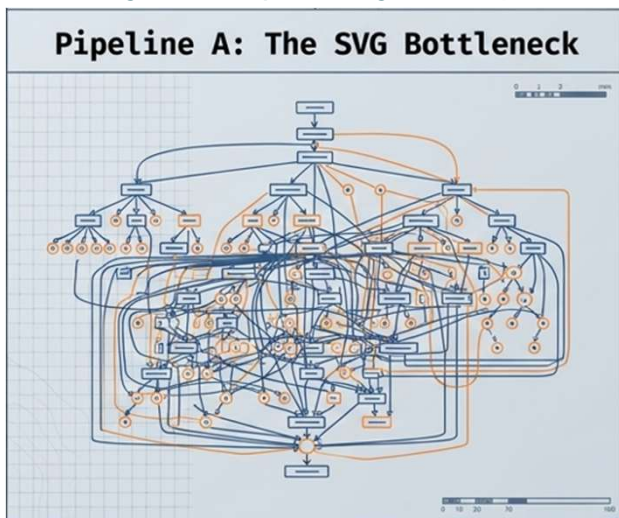


The reactivity loop

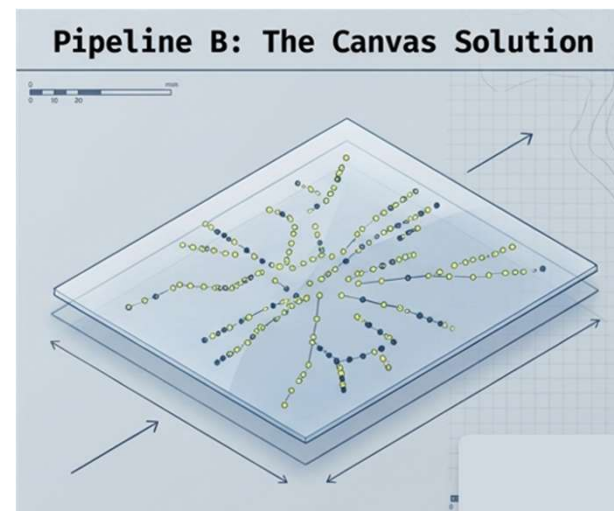


Performance optimization I : DOM vs. Canvas

- Standard rendering injects every coordinate as a distinct DOM node. At 50 000 points, browser memory exhausts, causing severe panning latency.

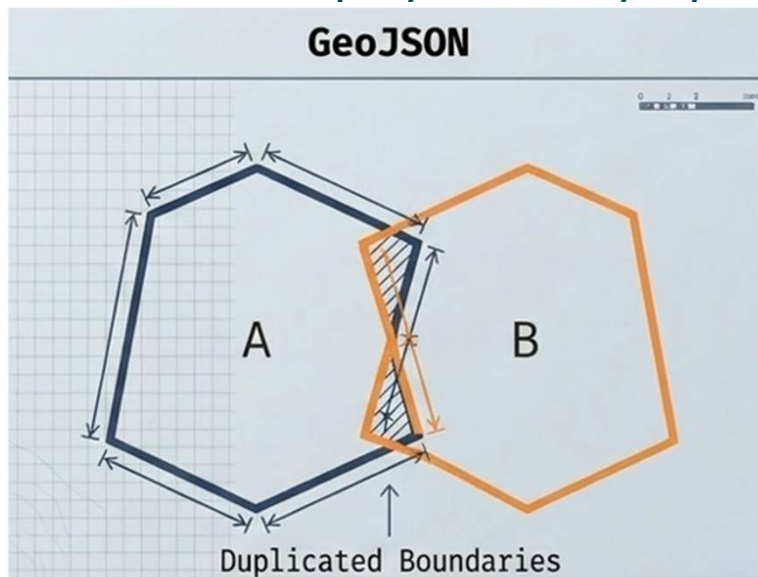


- Passing `prefer_canvas=True` forces Leaflet to draw all vector elements onto a single, flat pixel canvas, bypassing DOM complexity entirely.

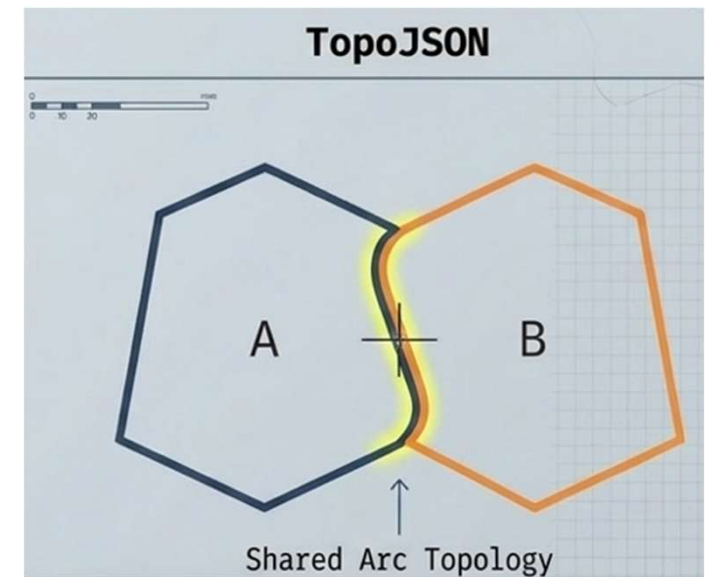


Performance optimization II: TopoJSON & Topology

- Stitching features via a shared index and applying quantized delta-encoding reduces file payloads by up to 80%



$$x_i^{encoded} = \text{round} \left(\frac{x_i - x_{i-1}}{scale} \right)$$



Performance optimization III : Data decimation

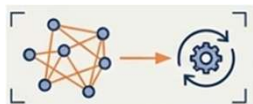
- Coordinate decimation
 - Limits precision to sub-meter accuracy ($p=6$), dropping 8 bytes per coordinate.

Before: `[-122.33207081567345, 47.60620951234567]` → After: `[-122.332071, 47.606210]`

- UUID Compression
 - Regex truncation of 32-character DOM UUIDs prevents HTML file bloat.

Before: `geo_json_a19eff2648beb3d74760dc0ddb58a73d` → After: `geo_json_a19eff26`

- FastMarkerCluster
 - Bypasses individual Marker instantiations. Writes a raw array of tuples directly into a client-side execution loop.

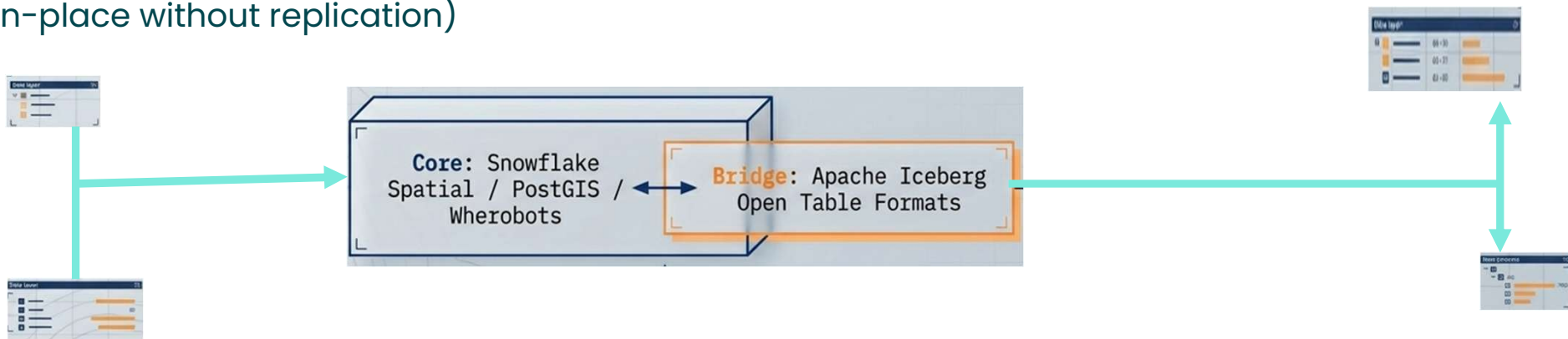


Modern Spatial Architectures

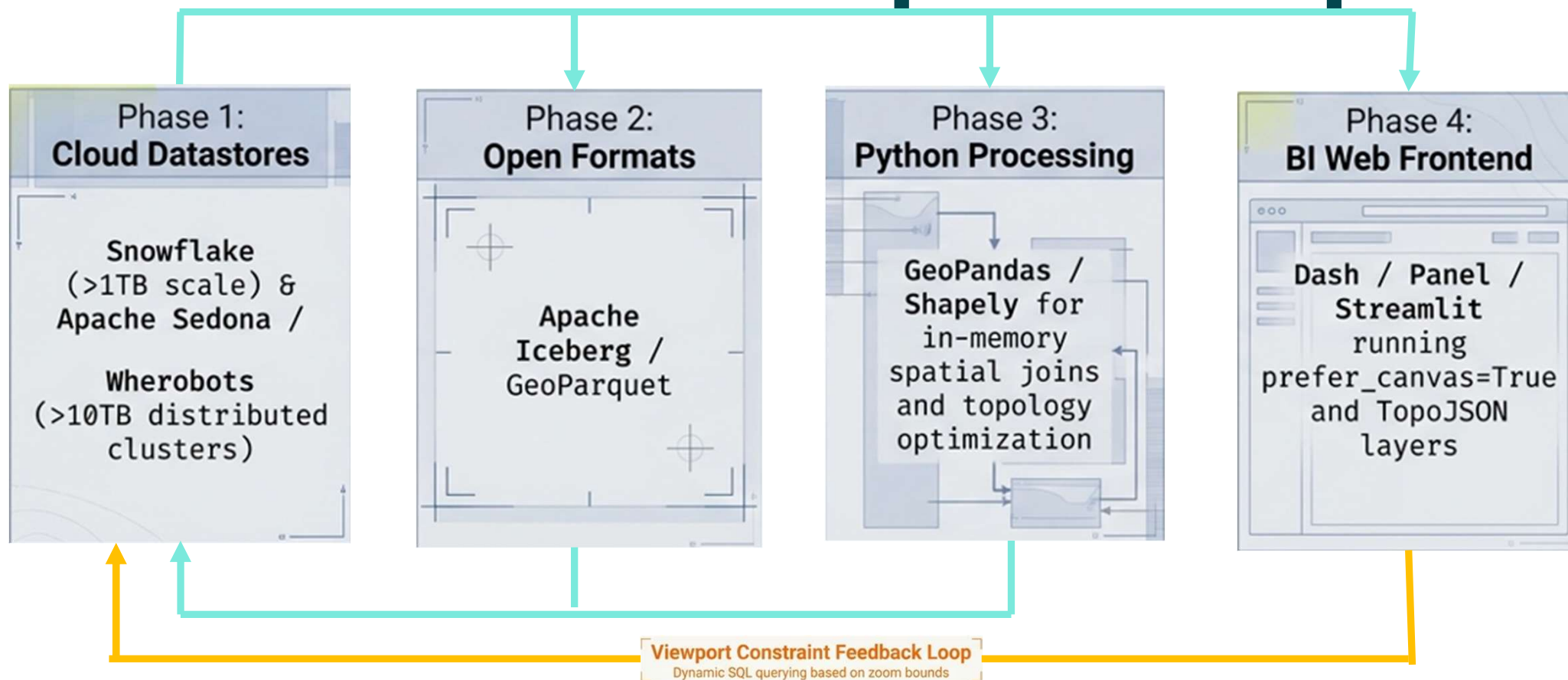
- Legacy Architecture (High latency, duplicate storage)



- Spatial Lakehouse (Result: Operational databases and analytical engines query the exact same data in-place without replication)



The Enterprise Blueprint





Consortium



Ascalia



diadikasia
BRIGHT GROUP
MEMBER



smartsense

Megasense



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Thank You!



Name: doc. dr.sc. Silvije Davila



Organization: Institute for Medical Research and
Occupational Health



Email: sdavila@imi.hr



Website: www.imi.hr



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